

◆ WEEK 2 — SESSION 4

MOCK EXAM — COMPUTE LAYER (EC2 + INSTANCE TYPES + AWS LAMBDA)

◆ CLOUD SECURITY FOUNDATION BOOTCAMP

◆ EXAM STRUCTURE

◆ TOTAL MARKS

100 Marks

◆ SECTIONS

Section	Type	Marks
Section A	Multiple Choice Questions	30
Section B	Short Answer Questions	30
Section C	Scenario-Based Questions	40

◆ SECTION A — MULTIPLE CHOICE QUESTIONS (2 MARKS EACH)

1. What does “compute” mean in cloud computing?

- A. Data encryption
- B. Processing power used to run applications
- C. Network routing
- D. User authentication

✓ Correct Answer: **B**

2. What does EC2 stand for?

- A. Elastic Compute Cloud
- B. Enterprise Compute Cluster
- C. Elastic Cloud Configuration
- D. External Compute Core

✓ Correct Answer: **A**

3. Which AWS service provides virtual servers?

- A. Lambda
- B. S3
- C. EC2
- D. CloudFront

✓ Correct Answer: **C**

4. Why is EC2 called “Elastic”?

- A. It compresses storage automatically
- B. Resources can scale up or down
- C. It creates backups automatically
- D. It encrypts traffic dynamically

✓ Correct Answer: **B**

5. Which instance family is best for high CPU workloads?

- A. Storage Optimized
- B. Memory Optimized
- C. Compute Optimized
- D. General Purpose

✓ Correct Answer: **C**

6. Which instance type is most suitable for AI workloads?

- A. GPU Instances
- B. General Purpose
- C. Storage Optimized
- D. Burst Instances

✓ Correct Answer: **A**

7. What is the purpose of a Security Group?

- A. Encrypt files
- B. Control network traffic
- C. Create IAM users
- D. Manage billing

✓ Correct Answer: **B**

8. What risk exists if port 22 is publicly open?

- A. Increased storage cost
- B. Public SSH exposure
- C. Slower application speed
- D. Database corruption

✓ Correct Answer: **B**

9. What is AWS Lambda?

- A. A database service
- B. A storage service
- C. A serverless compute service
- D. A monitoring platform

✓ Correct Answer: **C**

10. What does “serverless” mean?

- A. No servers exist
- B. AWS manages the infrastructure
- C. Applications do not need code
- D. Servers are free

✓ Correct Answer: **B**

11. Which workload is BEST for Lambda?

- A. Long-running enterprise ERP system
- B. Continuous gaming server
- C. Event-triggered automation
- D. Persistent database cluster

✓ Correct Answer: **C**

12. Which service gives full operating system access?

- A. Lambda
- B. EC2
- C. S3
- D. DynamoDB

✓ Correct Answer: **B**

13. Which AWS service automatically scales event-based workloads?

- A. EC2 only
- B. IAM
- C. Lambda
- D. Route 53

✓ Correct Answer: **C**

14. Why are instance types important?

- A. They determine compute resources
- B. They configure IAM automatically
- C. They manage DNS records
- D. They create backups

✓ Correct Answer: **A**

15. What is overprovisioning?

- A. Insufficient compute resources
- B. Using more resources than needed
- C. Encrypting storage incorrectly
- D. Restricting user permissions excessively

✓ Correct Answer: **B**

◆ SECTION B — SHORT ANSWER QUESTIONS

(5 MARKS EACH)

1. Explain the difference between EC2 and AWS Lambda.

2. Why are Security Groups important in AWS?

3. What are the benefits of serverless computing?

4. Explain why choosing the correct instance type matters.

5. What happens if infrastructure is exposed publicly without restrictions?

6. Why should businesses avoid unnecessary open ports?

◆ SECTION C — SCENARIO-BASED QUESTIONS

(10 MARKS EACH)

QUESTION 1 — INTERNET EXPOSURE RISK

A student launches an EC2 instance and opens:

- Port 22
- Port 80
- Port 443
- All other ports to the public internet

Explain:

- ◆ What security risks now exist
 - ◆ Which ports should remain open
 - ◆ Why unrestricted exposure is dangerous
-

QUESTION 2 — INSTANCE TYPE DECISION

A company is deploying:

- A machine learning platform
- A small internal HR website
- A large analytics database

Match each workload to the correct instance family and explain why.

QUESTION 3 — SERVERLESS ARCHITECTURE

A company wants to automatically resize uploaded images whenever users upload files to cloud storage.

Explain:

- ◆ Why Lambda is suitable
 - ◆ How event-driven execution works
 - ◆ Why EC2 may not be the best option
-

QUESTION 4 — GOVERNANCE THINKING

An engineer launches multiple large EC2 instances without approval.

Explain:

- ◆ Cost risks
 - ◆ Security risks
 - ◆ Governance failures
 - ◆ How organizations should control compute usage
-

◆ BONUS CHALLENGE QUESTION

(OPTIONAL)

“Modern cloud architecture is about using the right compute model for the right workload.”

Explain this statement using:

- ◆ EC2
 - ◆ Lambda
 - ◆ Security
 - ◆ Cost optimization
 - ◆ Scalability
-

◆ PRACTICAL MOCK LAB EXAM

Students must demonstrate:

- ✓ Launch EC2 instance
 - ✓ Configure Security Group correctly
 - ✓ Connect to instance
 - ✓ Create Lambda function
 - ✓ Execute test event
 - ✓ Explain EC2 vs Lambda differences
-

◆ PRACTICAL VALIDATION RUBRIC

Skill	Pass Requirement
EC2 Deployment	Instance launches successfully
Security Groups	Only required ports open

Skill	Pass Requirement
Lambda Deployment	Function executes successfully
Governance Thinking	Student explains risks correctly
Infrastructure Awareness	Student understands exposure

◆ FINAL EXAM PRINCIPLE

“Cloud engineers are not judged by how many services they launch — they are judged by how securely and intentionally they design systems.”